AP PRECALCULUS

Unit 1 • Worksheet 1.12MEGA

Worksheet 1.12MEGA • Transformations of Functions

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$g(x) = f(x-2)+5$. Describe the transformation.

(A) left 2, up 5
(B) right 2, up 5
(C) right 5, up 2
(D) left 5, down 2

Question 2

NO CALCULATOR

$g(x) = \frac{1}{3}f(x)$. Describe the transformation.

(A) vertical dilation by $\frac{1}{3}$
(B) vertical dilation by 3
(C) horizontal dilation by $\frac{1}{3}$
(D) horizontal dilation by 3

Question 3

NO CALCULATOR

$g(x) = f\left(\frac{1}{2}x\right)$. Describe the transformation.

(A) horizontal dilation by $\frac{1}{2}$
(B) vertical dilation by 2
(C) horizontal dilation by 2
(D) vertical dilation by $\frac{1}{2}$

Question 4

NO CALCULATOR

Which function is the graph of $y = |x|$ reflected over the x-axis and shifted right 3?

(A) $-|x+3|$
(B) $|-x-3|$
(C) $-|x|+3$
(D) $-|x-3|$

Question 5

NO CALCULATOR

The vertex of $y = x^2$ moves to $(-1, 4)$ and the graph opens downward with the same shape. Which is the equation?

(A) $y = -(x+1)^2 + 4$
(B) $y = -(x-1)^2 + 4$
(C) $y = (x+1)^2 - 4$
(D) $y = -(x+4)^2 - 1$

Question 6

NO CALCULATOR

$(6, 2)$ is on the graph of f. Which point is on $y = f(3x)$?

(A) $(18, 2)$
(B) $(6, 6)$
(C) $(2, 2)$
(D) $(2, 6)$

Question 7

NO CALCULATOR

f has range $[1, 5]$. What is the range of $y = -f(x) + 2$?

(A) $[-3, 1]$
(B) $[3, 7]$
(C) $[-5, -1]$
(D) $[1, 5]$

Question 8

NO CALCULATOR

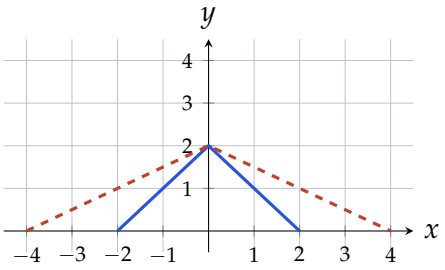
Which transformation can change the domain of a function?

(A) $f(x) + 4$
(B) $3f(x)$
(C) $-f(x)$
(D) $f(x - 4)$

Question 9

NO CALCULATOR

The graph of f is shown (solid). The dashed graph is g. Which is g?



(A) $f(2x)$
(B) $2f(x)$
(C) $f\left(\frac{1}{2}x\right)$
(D) $\frac{1}{2}f(x)$

Question 10

NO CALCULATOR

Which describes $g(x) = f(-x)$ for an even function f?

(A) $g = -f$
(B) $g = f$
(C) g is f shifted left
(D) g is undefined

Question 11 (Free response)

NO CALCULATOR

Let $f(x) = x^2$ and $g(x) = -2(x-1)^2 + 8$.

(a) Describe the transformations from f to g.
(b) Find the zeros of g and its maximum value.
(c) If f is restricted to $[-2, 2]$, find the corresponding domain and range of g.

Question 12 (Free response)

NO CALCULATOR

The graph of f is the line segments through $(-2, 1)$, $(0, 3)$, $(2, -1)$. List the corresponding points on each graph: (a) $y = f(x) + 2$ (b) $y = f(x + 2)$ (c) $y = -f(x)$ (d) $y = 2f\left(\frac{x}{2}\right)$